

S1 Supplementary Methods

Measurement contract, feature extraction, geometry, and audit protocol

Short title: S1 Methods - NeuralManifoldDynamics

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S1.0 Dataset and measurement contract

NeuralManifoldDynamics v3.0.0 defines an ingest-layer measurement contract rather than a single-dataset clinical experiment. The reference pair is EEG dataset ds004511 and fMRI dataset ds000228; validation examples include ds003838 (EEG, ECG, PPG, and pupil), ds003645 (MEEG), ds005555 (sleep), and ds006848 (verbal working memory). Two rodent DANDI dandisets extend coverage to spike-sorted and continuous electrophysiology: 000006 and 000009 (Neuropixels Units) and 000458 (Neuropixels LFP). Dataset-specific source, DOI, sampling-rate, channel, and task metadata are resolved by the YAML overlay used for each run. For example, the ds003838 overlay identifies OpenNeuro version 1.0.6 and DOI 10.18112/openneuro.ds003838.v1.0.6.

The reference EEG configuration writes an 8 s window every 4 s after preprocessing to 250 Hz, with a 1–45 Hz bandpass and average rereferencing. Other modalities use their own configured sampling and preprocessing policies. All downstream numbers must therefore be read against the selected run manifest, configuration, and feature table, not as direct computations on unprocessed raw tables. The package version is 3.0.0; optional phase-anchor, sleep-EAP, BDF, and NWB/ephys/LFP surfaces are described with their dataset-specific validation limits in S6.

Version 3.0.0 keeps the canonical [m,d,e] chart, stratified 9D names, and configured projection unchanged. It adds a versioned, fail-closed namespace for dynamical-family objects: canonical writes use `/dynamical_families/{diffusion,destination,resilience}/v1`, while the legacy `/orthogonal_dynamics/` surface is read-only. Family generation is disabled by default in common EEG, fMRI, and ephys profiles. The diffusion family measures local increment covariance; ingest always exports `drift=None`, so `A_bD` and `R_b_over_a` remain `not_testable` on this path. Computation status, measurement validity, grain, provenance, and `claim_status=no_biological_claim` travel with each computed family. Coordinate support and file-level capabilities are recorded separately at `/support_signature/v1(mndm.support_signature.v1)`.

S1.1 Epoch structure and feature export

The pipeline first creates a time-indexed feature surface and then applies the release-bound projection. The canonical and auxiliary outputs are:

- raw and strict robust-z feature surfaces, plus the transform-aware `features_projection_z` surface when configured;
- `coords_9d/values` with fixed subcoordinate names and derived `mnp3_3d = [m, d, e]`;

- `mnps_3d_dot`, window start/end, coverage, and geometry-validity metadata;
- global and optional regional Jacobians, including EEG channel groups and fMRI networks where support is sufficient;
- `feature_anchors` and explicit subject-anchored or cohort-anchored coordinate layers;
- `anchor_state`, `anchor_state_dot`, `anchor_quality`, and optional `anchor_coupling`;
- event-locked and block-native sidecars, which are derived layers rather than replacements for the canonical subject HDF5 measurement object;
- `row_source` provenance for simultaneous MEEG recordings.
- gated `dynamical_families/{diffusion,destination,resilience}/v1` objects, when explicitly enabled, with their support and claim metadata.

For the ds003838 block-native profile, 8 s windows are stepped by 4 s across task-state-derived sustained blocks. The sidecar retains the source-window index and task label so that a downstream analysis can distinguish window position from the canonical trajectory itself.

S1.2 Primary metric definitions

Let Y_t denote the modality-specific signal in a window and $z_t = \text{phi}(Y)_t$ the exported feature vector. The stratified chart is computed as

$$x_t^9 = W_{9D} z_t$$

and the canonical chart is the fixed weighted projection

$$x_t = P x_t^9 = [m_t, d_t, e_t]$$

.

The columns of P are normalized at runtime and the implementation renormalizes over finite supported weights when an axis is partially observed. The resulting coverage is retained; a degraded-support coordinate is not silently treated as identical to a full-support coordinate. The

nine subcoordinates are grouped as (m_a, m_e, m_o), (d_n, d_l, d_s), and (e_e, e_s, e_m).

They are release-fixed operational proxy families, not uniquely identified biological axes.

The time derivative `mnp3d_dot` uses Savitzky–Golay smoothing on the epoch series. The common EEG defaults are window 7 and polynomial order 3; the fMRI reference uses window 5 and order 2. Short sequences fall back to central differences, and robust segmentation avoids smoothing across large jumps.

When support is adequate, the local dynamical layer estimates a chart-level Jacobian

$$J(x_t, t) = \frac{\partial f(x, t)}{\partial x} \Big|_{x=x_t}$$

.

The common reference uses a Euclidean k-nearest-neighbour fit with $k = 20$, ridge regularization $\alpha = 1$, whitening, and a three-window super-window. The exported Jacobian is an empirical local summary of the chart trajectory; it is not direct access to a unique biophysical state equation. Frobenius norm, trace-like summaries, condition number, and anisotropy describe magnitude, local contraction/expansion, conditioning, and directional imbalance. These summaries are interpreted only inside the recorded validity domain.

The embodied surface is parallel to, not part of, the canonical chart. HRV v0.1 provides HR mean, IBI mean, SDNN, RMSSD, pNN50, NN count, artifact fraction, coverage fraction, and quality score. The anchor-state hierarchy uses available physiological sources to derive `vagal_index`, `sympathetic_index`, `anchor_index`, and `vascular_index`; source choice and quality remain part of the provenance.

S1.3 Auxiliary feature extraction

ECG features are extracted from detected R peaks and may use centered 60 s HRV superwindows. PPG contributes rate, amplitude, variability, and quality features. Pupillometry

contributes diameter, volatility, blink-rate proxy, and quality features when a pupil trace is available. These are optional modality surfaces and are not required for the canonical MNPS chart.

MEG extraction produces separate magnetometer, gradiometer, and combined `meg_*` columns. These columns are shadow-mapped through the existing 9D contract. For physical MEG power, `features_projection_z` applies the configured `log10`, `robust-z`, and clipping sequence before export; raw-space `robust-z` is not an adequate substitute for those columns.

Neuropixels/ecephys extraction covers two source types with two different routes into the contract. NWB `Units.spike_times` are histogram-binned per unit into a population firing-rate matrix (default bin width 0.05 s, i.e. a 20 Hz rate grid, with an optional 0.05 s Gaussian smoothing kernel), then windowed (default 2.0 s window, 1.0 s step) into `ephys_*` feature columns (`ephys_mean_rate_hz`, `ephys_rate_std_hz`, `ephys_population_cv`, `ephys_participation_ratio`, `ephys_top_pc_variance_fraction`, `ephys_pairwise_corr_mean`, `ephys_rate_entropy`, `ephys_ar1`, `ephys_active_unit_fraction`). These fill `coords_9d` through a dedicated ephys-specific subcoordinate map, not through MEG-style shadow mapping onto `eeg_*` slots. NWB `ElectricalSeries LFP` channels instead route through the existing EEG feature and 9D contract with rodent-adapted band edges and optional depth-balanced contact selection; the `Units` path is disabled for these runs. Both ephys paths are validated only as transport-and-descriptive QC on rodent dandisets, with no readiness gate analogous to the MEG pilot; see S3.4d.

The optional `phase_anchor` extractor and `sleep-EAP` modules are present in the post-v2.4 working tree, but are not release-bound v2.4.0 methods. Event-locked and block-native exports are derived post-processing layers with their own detector, window, join, and QC provenance. Conventional EEG comparator packs are similarly additive and do not alter the MNPS projection.

S1.4 Physiological confound and secondary feature definitions

The principal physiological controls are HRV artifact and coverage fractions, ECG polarity, R-peak count, task-event overlap, PPG quality, pupil quality, EOG blink rate, and high-frequency power used as an EMG proxy. HRV superwindows are not assumed to be phase-pure: in ds006848, 87.7% of verbal working-memory windows overlapped retrieval events, so working-memory HRV claims were withheld. This is a dataset-specific gate, not a universal contamination percentage.

Temporal controls include epoch count, available seconds, minimum direct-axis coverage, source-window matching, block position, event overlap, and derivative segmentation. Geometry controls include non-finite coordinates, duplicate subcoordinates, condition number, anisotropy, and minimum support for local fits. The `standard_invalid_v1` policy records why a requested geometry or Jacobian was withheld.

Secondary features include conventional spectral ratios, Hjorth measures, permutation or sample entropy, regional summaries, inter-network `coupl_*` columns, and the additive anchor indices. They are not automatically primary endpoints; their role is fixed by the named validation package or analysis script.

S1.5 Statistical design

This manuscript has two analysis tiers. Contract QA uses run manifests, coverage, validity, schema, and reproducibility hashes. Cohort validation analyses aggregate windows within subject/session before testing condition or stage contrasts. The ds003838 statistics package used subject-level stage medians, a Friedman repeated-measures test across rest/listen/mem5/mem9/mem13, pairwise Wilcoxon signed-rank tests, and Cohen's **d**. Its complete-case inferential sample was $N = 62$; the separate full export contained 130 HDF5 subjects.

BH-FDR correction is used inside analysis families where the cited claims ledger specifies it, such as the ds006848 pairwise encoding contrasts. The manuscript does not define one article-wide global coordinate FDR family. Accordingly, no global null or universal alpha threshold is invented here; each supplementary result must name its analysis package, family, correction, and sample. Descriptive completed-run counts are not inferential sample sizes.

S1.6 Robustness and confound audit protocol

The available audits are mapped to the contract as follows:

1. **Coverage and intersection:** report retained epochs, seconds, axis support, and the complete-case sample for each analysis.
2. **Geometry validity:** apply `standard_invalid_v1`, including finite coordinates, duplicate checks, conditioning, and Jacobian support.
3. **Time-grid and derivative QA:** test epoch times, jumps, segmentation, and derivative behavior before local fits.
4. **Reproducibility:** compare saved feature, neighbour-index, coordinate, and Jacobian hashes across replayed worker counts where available.
5. **Within-subject validation:** use subject-level aggregation and permutation or label-shuffle nulls when the named analysis package supplies them.
6. **LOOSO and bootstrap:** report only where the relevant script produced an output; ds006848 has cohort-specific leave-one-subject-out and bootstrap checks, not a universal MNDM guarantee.
7. **Physiology:** inspect polarity, HRV artifact/coverage, event overlap, anchor quality, and source selection.

8. **Artifact balance:** compare EOG/high-frequency and conventional features; note that ICA was not run for the ds006848 claim package.
9. **Modality-specific safeguards:** use `row_source` and `features_projection_z` for MEG, and `channel/contact/reference QC` for DANDI LFP.
10. **Derived-layer sensitivity:** keep event-locked and block-native window profiles, source indices, and matching diagnostics visible.

These audits are cohort-specific. A passed QC gate does not establish cross-modal robustness or chart invariance.

S1.7 v3.0 dynamical-family contract and analysis boundary

The v3 family layer is a measurement surface beside the chart, not a new coordinate ontology. The common configuration leaves it off unless a named profile requests it. Diffusion uses local increment covariance with the configured neighbourhood and validity rules; it does not silently replace the chart's derivative, add a ninth or tenth canonical axis, or infer an Itô drift. Analysis packages may estimate chart velocities, compare local estimators, and define null controls, but those choices remain outside ingest and must retain their own cross-fitting, support, and provenance metadata.

The production contract therefore distinguishes a computed family object from a scientifically qualified dynamical claim. Missing independent velocity is represented as `drift=None` and the associated alignment fields are withheld, not filled with a surrogate or zero. Every family object carries the explicit `no_biological_claim` ceiling.

The container rename does not rename schema identities. Diffusion, destination, and resilience retain `mndm.diffusion_geometry.v1`, `mndm.committor.v1`, and

`mndm.finite_amplitude_resilience.v1`, respectively. The separate opt-in reachability object retains `mndm.stochastic_reachability.v1` under `/stochastic_reachability/v1`; it is not a `dynamical_families/spread` YAML family and its `w_Q` is discrete transition spread rather than `a_hat` or $\hat{a}$ process noise.

S1.8 Bounded CHART and M3 analysis methods

The bounded CHART audit tests coordinate behavior under controlled transforms. Orthogonal and isotropic transforms preserve the audited local quantities to a maximum discrepancy of 2.9×10^{-14} in the fixed synthetic check. Anisotropic transforms are expected to change neighbourhood geometry and local estimates; that sensitivity is evidence about the estimator, not a failed invariant. Projection from 3D to an observed subspace loses hidden-coordinate information. The audited invariants are the transformed vectors and covariances under orthogonal and isotropic transforms; anisotropic sensitivity is expected. The scalar CHART diagnostic is unregularized and is not a production C1 qualification of `A_bd`.

The bounded M3 extension uses six synthetic systems: full linear and nonlinear 3D, full linear and nonlinear 9D, linear latent 9D observed through the first three coordinates, and noisy linear 3D. Linear transitions use $A = 0.65 I + 0.08 S$, $Q = 0.09 I$, and $dt = 0.1$; nonlinear transitions add $0.12 \tanh(Sx)$, where $Sx = \text{roll}(x, 1)$ is the fixed cyclic shift. Linear paths are initialized from the stationary Gaussian covariance and nonlinear paths from zero; both use a 300-transition burn-in before the 1800 retained training states. The partial and noisy systems use the observed one-step conditional target; they do not claim to recover a latent drift. Query paths use seeds 9001–9003. For the noisy case, $R = 0.09 I$. With stationary covariance P , observation matrix H , and observation noise R , the recorded analytic quantities are $P = A P A^T + Q$, $V =$

$H P H^T + R$, $C = H A P H^T$, and $b_{\text{obs}}(y) = (C V^{-1} y - y) / dt$. With $C_2 = H A^2 P H^T$, the lag-memory witness is the Frobenius norm of $C_2 - C V^{-1} C$, while the conditional next-state covariance is retained separately as $V - C V^{-1} C^T$. These are model-derived Gaussian diagnostics, not estimator or qualification gates.

There are twelve independent training fits (seeds 101–112) and three fixed held-out query trajectories of 49 states each, giving 144 genuine query sources. Three zero-padding states separate the training and held-out segments; padding, seams, and artificial carrier increments are excluded from fit and evaluation. Local-mean and local-linear cross-fitting use $k = 64$, minimum support 16, and a temporal embargo of $2 dt$. Comparators share the same raw evaluation-neighbour covariance and covariance scaling. The oracle uses the same evaluation neighbours. The negative control is named **cross-segment order corruption**: a roll of 17 changes the mapping while keeping truth fixed; it is not a temporal-shift claim.

The extension records per-query predictions and support, pointwise bias and population variance across the twelve fits, sample variance and descriptive MCSE, and the MSE identity where support is complete. The analytic observed conditional covariance is retained separately from the raw neighbourhood covariance. The partial-system memory witness (6.58×10^{-9}) and noisy-system witness (0.0441) are model-derived Gaussian diagnostics; they do not establish Markov sufficiency, production ingest qualification, biological validity, or a universal estimator ranking. The integrated suite contained 117 tests, including 18 dedicated extension tests.

S1.9 Secondary analysis

The optional `anchor_coupling` surface provides body–brain and inter-network coupling summaries. A completed article-level EAP partial-correlation family with a locked number of tests

and corrected table is not established in the v2.4 evidence pack. It is therefore an exploratory analysis surface, not a filled primary result. Any future coupling analysis must predeclare the aggregation unit, covariates, missingness rule, test family, correction, and run identifier.

Bibliography

Competing Interests

The authors have declared that no competing interests exist.